

GAME DESIGN DOCUMENT

Abyssal Jump

“The caverns remember every journey”

1. Game Overview

1.1. Game Summary

Title: Abyssal Jump

Genre: 2D Precision Platformer

Platform: PC (Windows)

Target Audience: Players aged 8+, fans of fantasy platformers

Development Team: 1 person

Development Time: 1 months

Engine: Unity

Abyssal Jump is a fast-paced 2D platformer centered around fluid movement, precision, and mastery. Players must navigate treacherous caverns and ascend toward the exit as quickly as possible. Along the way, they can unlock powerful magical abilities that open up new ways to traverse the environment and overcome obstacles.

1.2. Design Motivation

I've always enjoyed the sense of progression found in Metroidvania games, and I wanted to combine that feeling of discovery and growth with the precision and challenge of a platformer. In Abyssal Jump, progression is closely tied to mastery where players must overcome challenging obstacles to unlock new abilities, which then become essential tools for navigating later stages.

This creates a progression loop where mastering one challenge unlocks new possibilities, encouraging players to continuously improve their movement skills while discovering new ways to explore the caverns.

1.3. Design Pillars

The game is designed around five primary pillars:

1. Fluid Movement

Movement should feel responsive, consistent, and predictable, allowing players to build confidence and execute actions with precision.

2. Precision

Challenges should reward careful control and a strong understanding of the movement mechanics, encouraging players to refine their timing and positioning.

3. Mastery

Players should improve through repeated attempts, learning from mistakes and gradually becoming more efficient and consistent in their movement.

4. Exploration Through Abilities

New abilities should expand the player's movement options, revealing new traversal possibilities and allowing obstacles to be approached in different ways. Players can also discover secret passages and hidden routes by experimenting with their abilities and paying attention to the environment, rewarding curiosity and exploration.

5. Speed

A timer encourages players to optimize their movement and routes. Rather than being the initial focus, speed becomes a measure of mastery once players have learned the level and become familiar with its challenges.

2. Gameplay

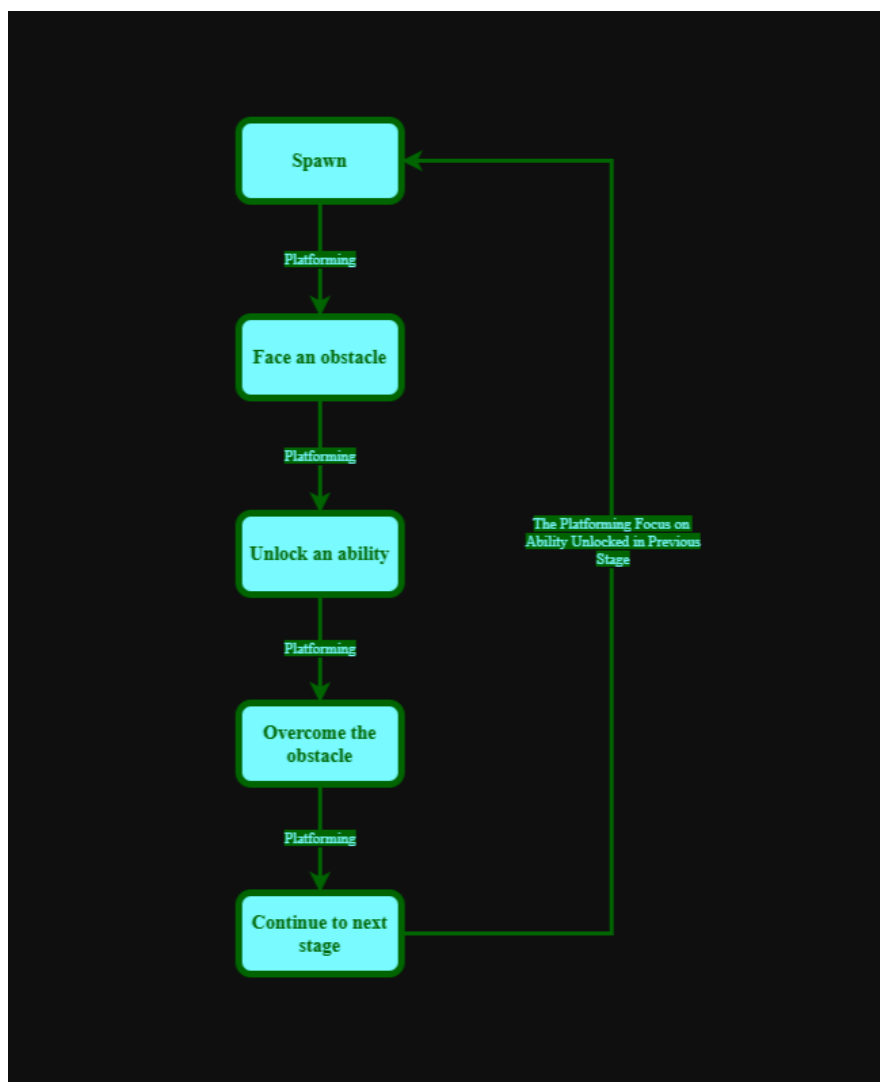
2.1. Gameplay Flow

In Abyssal Jump, players must navigate challenging, precision-based platforming sections to reach the end of each stage. Along the way, they will encounter obstacles that require them to unlock and use new abilities in order to progress. Once an ability is acquired, players can use it to overcome these obstacles and continue to the next stage.

A timer runs throughout the level, encouraging players to complete each stage as quickly as possible. Players are challenged to improve their movement, optimize their routes, and achieve the shortest completion time they can.

2.2. Core Gameplay Loop

The primary gameplay loop consists of:



3. Mechanics

Abyssal Jump features several mechanics:

3.1. Basic Movement

Players can move horizontally through each stage. This is the primary method of traversal and allows players to navigate the caverns. Basic movement provides players with direct control over their character and establishes the foundation for more advanced movement abilities.

3.2. Jump

Players can jump to reach higher platforms and traverse vertical sections of the caverns. Jumping serves as the foundation for the game's platforming and is essential for navigating gaps and obstacles.

3.3. Dash

Players can quickly dash horizontally in a chosen direction, allowing them to cross large gaps and reach otherwise inaccessible areas. The dash can also be used to pass through glowing thorn vines, clearing paths and allowing players to progress.

3.4. Double Jump

The Double Jump is an extension of the basic Jump ability. Players can perform an additional jump while airborne, providing greater vertical reach and more control over their movement.

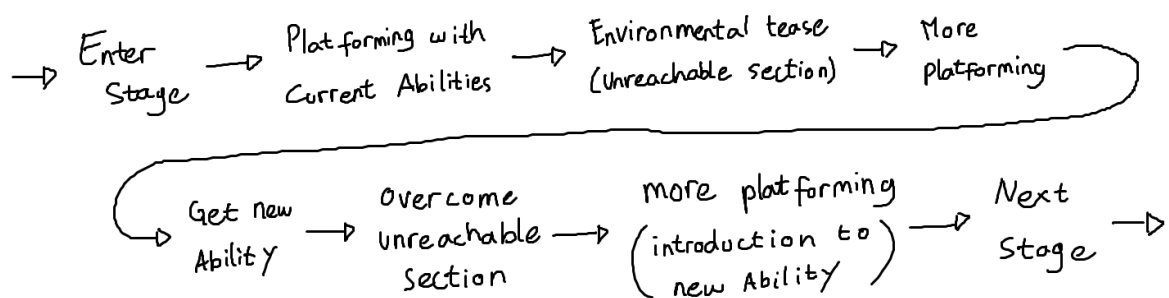
3.5. Omni-Dash

The Omni-Dash is an extension of the Dash ability that allows players to dash in all eight directions, allowing players to move horizontally, vertically, and diagonally. This provides greater freedom of movement and opens up new traversal possibilities, allowing players to navigate more complex obstacles and access previously unreachable paths.

4. Level and Progression

4.1. Level Design

Each level is designed with this flow in mind:



Players begin each stage with their currently available abilities and encounter platforming challenges designed around those abilities. As they progress, the environment introduces teases of unreachable areas, prompting players to wonder how they might access them.

Players continue exploring the stage while testing and improving their mastery of their current abilities. Eventually, they obtain a new ability that provides a way to overcome the previously unreachable section. This creates a sense of discovery and gives the new ability an immediate purpose.

After acquiring the ability, players are guided through a short section designed to introduce and reinforce its mechanics. Once players become familiar with the new ability, they continue toward the next stage, where the cycle of exploration, discovery, and mastery begins again.

4.2. Skips and Secret Passages

Throughout the stage, skips and secret passages are placed to encourage exploration and curiosity while testing the player's understanding of each ability. Discovering these hidden routes allows skilled players to reach the ending faster.

Dark shadows are used to conceal the entrances and keep the passages hidden. Once the player enters, the shadow disappears, revealing the secret route and rewarding their exploration.

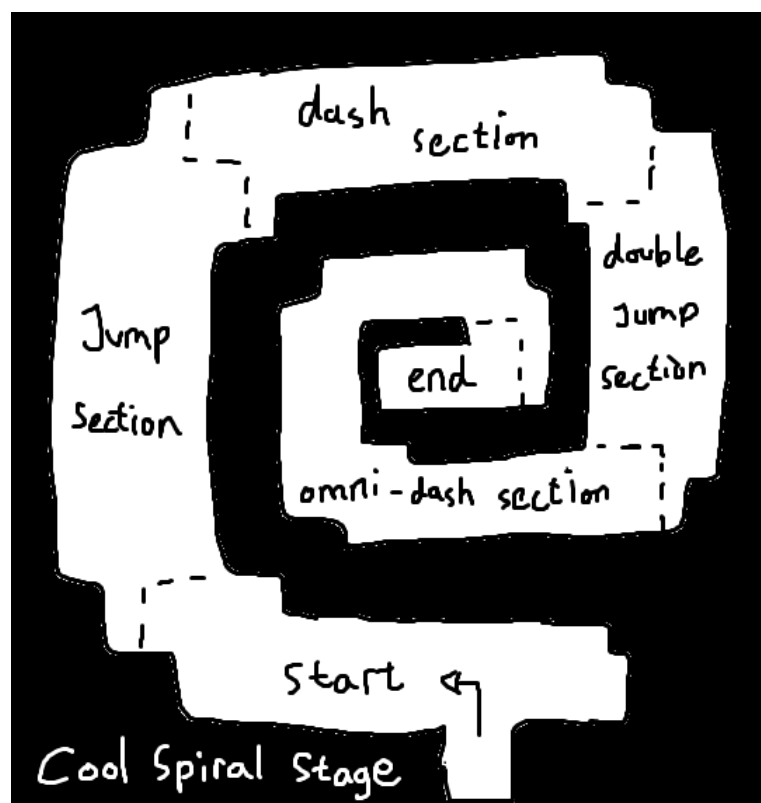
4.3. Progression

The player begins with basic horizontal movement and must traverse the caverns to obtain the Jump ability. Jump allows them to navigate the cavern's winding paths and reach previously inaccessible areas.

Next, the player unlocks Dash, allowing them to cross horizontal gaps and pass through glowing vine thorns blocking their path. The Double Jump follows, providing greater vertical reach and improved control while navigating the caverns.

Finally, the player unlocks Omni-Dash, which enables horizontal, vertical, and diagonal dashes. This ability is primarily used to navigate through sections filled with vine thorns positioned in various orientations, putting the player's understanding of all movement abilities to the test.

The final level is a long platforming sequence designed to test the player's mastery of all four abilities. It is divided into four main sections, with each section focusing on one of the abilities learned throughout the game.



Players must apply their skills and adapt to increasingly challenging platforming situations. By pushing through all four sections, they will eventually reach the ending, marking the culmination of their journey and mastery of the game's movement mechanics.

5. Time Trial and Scoring

Throughout the game, a timer continuously counts upward and stops when the player reaches the end. It serves primarily as an incentive for players to improve their completion time, rewarding those who master the level, optimize their movement, and discover more efficient routes without limiting players who prefer a more casual pace.

At the end of the stage, the player's completion time is displayed alongside a message encouraging them to retry and beat their previous time. The caverns remember every journey, making each replay an opportunity to improve, discover new routes, and achieve a faster completion time.

6. UI / UX

6.1. UI Design

The UI uses a simple monochromatic interface inspired by Hollow Knight, with a stylized font and minimal visual elements. This keeps the interface clean while maintaining the game's dark and atmospheric visual identity.

6.1. Main Menu

The main menu displays a cavern environment with glowing plants as visual points of emphasis, creating an immersive atmosphere and encouraging curiosity about the game world. Simple typography is used for the title and buttons to maintain a minimal yet atmospheric presentation.

6.2. Tutorial

Each ability has a dedicated tutorial that appears when the player acquires it, similar to the ability tutorials in Hollow Knight. The tutorial uses a simple ability icon accompanied by concise narrative text that explains the ability and how it can be used, allowing players to quickly understand their new movement option.

7. Art and Visual

7.1. Art Style Overview

The game uses a stylized 2D fantasy aesthetic inspired by Hollow Knight. The art direction focuses on dark cavern environments and a mysterious atmosphere, while maintaining clear visuals that keep character movement, platforms, and interactive elements easy to read during gameplay.

The game's visual assets were primarily sourced from Maaot 2D free asset pack from itch.io. These assets were selected and integrated based on their compatibility with the game's visual direction and gameplay requirements.

7.2. Art Direction

The art direction aims to achieve the following:

7.2.1. Readability

Players should be able to clearly distinguish between platforms, hazards, and interactive objects while maintaining clear visibility of the character and their movements.

7.2.2. Atmosphere

The cavern environments should convey a mysterious and dangerous atmosphere without compromising gameplay readability.

7.2.3. Consistency

Characters, environments, UI elements, and visual effects should follow a cohesive stylized visual language, maintaining the game's mysterious fantasy aesthetic.

7.3. Environment

The environments are intentionally kept dark to create a mysterious atmosphere. Lighting is used sparingly and concentrated around major platforming sections, helping guide the player's attention without overwhelming the scene.

Lights also highlight interactable objects and important areas, such as the main path and potential secret passages. Vibrant colors from the assets provide strong contrast against the dark environment, ensuring that important gameplay elements remain clearly readable while preserving the game's mysterious atmosphere.

8. Music and Audio

8.1. Audio Direction

The audio design aims to create a mysterious and immersive atmosphere that complements the dark cavern environment. Environmental sounds vary in volume and intensity to give each area a distinct sense of space, creating an immersive experience. Clear audio feedback is also used for player actions and movement. These sounds provide a responsive and satisfying platforming experience.

8.2. Music

The game music is sourced from nojisuma's royalty-free music gallery on Pixabay. The selected tracks convey a mysterious yet calming atmosphere, complementing the cavern environment while remaining subtle enough not to distract players from platforming.

The music uses spatial audio to enhance the sense of depth and immersion within the caverns. Environmental ambience is layered alongside the music to further establish the atmosphere and make each area feel more alive and immersive.

8.3. Sound Effects

All sound effects are sourced from Pixabay and are used to provide immediate feedback for player actions, making interactions feel more responsive and immersive. Important gameplay elements and player actions are given distinct sounds so players can easily recognize and respond to them.

Gameplay sound effects include:

- Walking, jumping, and landing.
- Dashing.
- Reaching checkpoints.
- Interacting with ability objects and obtaining new abilities.
- UI interactions, such as button clicks and menu navigation.

9. Playtesting and Iteration

I conducted several playtesting sessions for Abyssal Jump to evaluate the gameplay experience and identify areas for improvement. Based on feedback and observations from these sessions, I made several design changes to improve the overall player experience.

Problem	Observation	Design Change	Result
Dash movement feels difficult to control	Dash deceleration is important for vertical movement in Omni-Dash, but the current horizontal movement is difficult to control.	Reduce horizontal dash acceleration while preserving the existing deceleration needed for vertical traversal.	Dash feels easier to control horizontally while retaining its utility for vertical movement.
Some secret passages are too difficult and inconsistent	Some secret passages are intentionally tight and challenging, but their hitboxes are incorrectly positioned, making access inconsistent.	Adjust the hitboxes of affected secret passages.	Secret passages become more consistent and accessible while still requiring mastery.

10. Development Notes

10.1. Scope

Abyssal Jump is a solo project developed under a tight deadline. The scope was intentionally limited to a small number of movement abilities and stages, allowing each mechanic to receive sufficient development and testing. Enemies were also excluded from the game to avoid spending valuable development time on enemy design, behavior, and related UI systems.

10.1. Design Challenges

Throughout the development of Abyssal Jump, several challenges were encountered:

- Making movement feel responsive and satisfying.
- Designing levels that are both fun and challenging while being tailored to each ability.
- Making effective use of a limited number of visual assets.
- Establishing the game's atmosphere through lighting and audio.
- Balancing speedrun optimization with exploration through skips and secret passages.
- Creating an experience that is challenging and engaging without becoming frustrating.

10.2. Lessons Learned

Developing Abyssal Jump in approximately one month taught me that movement-based games require iteration at a smaller scale than initially expected. While the core movement mechanics can be established relatively quickly, much of the development effort goes into level and atmosphere design.

Creating platforming sections that are both challenging and enjoyable, while maintaining an immersive world, requires continuous refinement. Small adjustments to movement, level layouts, lighting, audio, and visual details can significantly affect how a level feels. This experience taught me the importance of perseverance, attention to detail, and repeated iteration in creating a cohesive and enjoyable game.